DAO CAO THAI

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PROFESSIONAL SUMMARY

Tech Lead and Backend & Data Engineer with 8 years of experience building high-load systems that demand strong data integrity. Serving as Tech Lead and Principal Developer across all core projects, spearheading the architecture while leading engineering teams of up to 10. Strong foundation in applied mathematics from quant finance, ML, and robotics. Growing in **Java** and **Go**, having designed production-grade codebases and shipped resilient applications in both, alongside expanding DevOps expertise and professional English communication.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Lua; Java, Go.
Backend & Data: FastAPI, Spring Boot, gRPC, WebSocket, PostgreSQL, SQL Server, Redis/Valkey, ClickHouse, DuckDB, Event Hubs, RabbitMQ, Avro, Parquet, PySpark, Airflow, NumPy/CuPy.
Cloud & DevOps: Azure AKS, AWS S3, GCP, Cloudflare Workers, Durable Objects, R2, KV, D1, MinIO, Docker, Kubernetes, FluxCD, OpenTofu.
Domains: AaaS: API Hub platforms with real-time usage-based billing, healthcare: HL7 FHIR R4, quantitative finance, Applying AI to the software development workflow.

PROFESSIONAL EXPERIENCE

Venera AI — *Tech Lead / Backend & Data Engineer* *2024 – 2026*

Venera API Hub *Tech Lead, team of 10*

Python, FastAPI, PostgreSQL, Valkey, Lua, gRPC, Azure Event Hubs, Avro, MinIO, ClickHouse, React, AKS.
OpenAI-style API platform for healthcare. In production, integrated with NanoSoft HIS.

- Separate the platform into a **Control Plane** and a **Data Plane** to isolate user management workloads from high-throughput serving API workloads. Use **gRPC** with single-flight pattern for low-latency inter-plane communication.
- **Control Plane:** Design database schemas for organizations, projects, fine-grained IAM, API keys, and billing reports. Build React dashboards for client project configuration and internal admin management (catalogs, pricing, entitlements, and credit ops).
- **Data Plane:** Implement **Valkey** as a low-latency caching layer to handle real-time authentication, entitlement checks, and credit validation before proxying streaming HTTP/WebSocket traffic to executors. Develop atomic billing workflows directly in Valkey using structured **Lua scripts**.
- **High-Throughput Hot Path:** Isolate the serving path from database bottlenecks (PostgreSQL/Event Hub) by operating purely on the caching layer. Stream usage metrics asynchronously from Valkey to **Azure Event Hub** (with PostgreSQL fallback for fault tolerance). Apply the **Transactional Outbox pattern** on the Control Plane to maintain real-time consistency between user configs and the cache.
- **Billing & Analytics Pipeline:** Forward usage logs to a MinIO Data Lake (Parquet format) and ingest into **ClickHouse** for analytics and billing reports. Use **Avro** with CRC-64 fingerprinting to manage schema evolution safely.

Internal ETL Platform *Tech Lead, team of 2*

Python, Airflow, DuckDB, PySpark/Delta Lake, Hugging Face

- Architected an **Airflow**-orchestrated ETL platform with a no-code UI, enabling non-technical users to build pipelines using minimal SQL knowledge.
- Streamlined data ingestion for the ML team by automating the crawling, processing, and loading of training-ready datasets into Hugging Face, SQL Server, and PostgreSQL for medical LLM initiatives.

Vietnamese FHIR Interoperability *Tech Lead, team of 4*

Java 21, Spring Boot, HAPI FHIR, FHIR Mapping Language (FML), FSH

- Authored **FHIR R4 Implementation Guides** aligning with MOH EMR and BHYT standards, establishing a national healthcare data interoperability baseline using **FSH**.
- Built Java services to transform BHYT XML into FHIR via **FML StructureMaps**, validating and storing resources in a **HAPI FHIR** server exposed via an API Hub for hospital integration.

Weegroup — *Backend Developer* *2025 – 2026*

TypeScript, Cloudflare Workers/Durable Objects/KV/R2/D1, TON Connect, RabbitMQ, Telegram Mini Apps

- Architected a serverless, low-latency backend on **Cloudflare Edge** (Workers, itty-router) for Telegram Mini Apps, serving nearly 1M global users with zero region-bound server overhead.
- Designed single-writer game states using **Durable Objects** to ensure strong consistency without database locks; leveraged **KV** for low-latency global reads, **R2** for media storage, and **D1** for analytical dashboards.
- Integrated **TON Connect** and Telegram Mini App authentication to power seamless Web3 payment flows, handling in-game item purchases, deposits, and TON token transactions.
- Implemented an asynchronous withdrawal pipeline using **RabbitMQ** to decouple slow on-chain transaction finality from real-time gameplay, enabling safe retries and fault isolation.

6P Capital Management — *Quantitative Developer & Data Engineer* *2021 – 2025*

Python, NumPy, CuPy, Pandas, Mosek, SQL Server, Airflow, FastAPI, WebSocket, Dash, React, NestJS

- Engineered an end-to-end quantitative research platform for Vietnamese equities (team of 3), orchestrating real-time streaming data ingestion from **SSI and VPS WebSockets**, Airflow ETL pipelines, and daily backtest-ready data marts.
- Designed high-performance **SQL Server schemas** to store complex simulation outputs, including alpha signals, portfolio positions, multi-period backtest metrics, and cross-sectional performance.
- Developed a **WorldQuant BRAIN-style DSL (Domain-Specific Language)** enabling researchers to express alpha logic via operators instead of raw code; implemented underlying operators using GPU-accelerated **NumPy/CuPy** with highly optimized rolling-window time-series functions.
- Built modular **simulation engines** supporting sector-neutral, group-neutral, and liquidity-constrained backtests, computing institutional-grade metrics (Sharpe, fitness, turnover, max drawdown, and benchmark PnL vs. VN-Index).

- Integrated the **Mosek convex optimization solver** into an alpha combination engine with pre-built graph-node workflows, allowing researchers to optimize multi-alpha portfolios without manual mathematical modeling.
- Developed full-stack research dashboards (React, Dash, NestJS) for alpha discovery and PnL visualization; researched and prototyped **Barra-style multi-factor risk models** for portfolio risk decomposition.

VMO Holdings — *Backend & Data Engineer* *2019 – 2021*

Python, Apache Airflow, PipelineWise, Airbyte, Apache Spark, Docker

- Built a self-service Data Integration Platform for Setel (Malaysian fintech) as part of an 8-engineer data team, transforming manual file-based configurations into a fully API-driven managed service.
- Wrapped **PipelineWise** and **Airbyte** execution engines into unified RESTful APIs, enabling dynamic programmatic creation, configuration, and management of data extractors and loaders.
- Engineered a metadata-driven **Airflow** orchestration layer that accepts pipeline definitions as JSON payloads and automatically generates dynamic DAGs, eliminating hand-written Python DAG files.
- Exposed unified endpoints for on-demand and scheduled execution, powering an internal management UI that allowed non-technical users to onboard new data sources with zero code changes.

AI-ASSISTED ENGINEERING

hitl — Human-in-the-loop Spec-Driven Development CLI *Personal Project, 2026*

- Designed an orchestration plugin for Claude Code featuring 17 custom skills, 7 isolated sub-agents, and automated enforcement hooks to enforce strict spec-driven development workflows.
- Implemented a Node.js **AST indexing engine** to lock human-approved API contracts, detect codebase drift, and maintain end-to-end requirement traceability to test suites.

EDUCATION & CORE COMPETENCIES

Foreign Trade University — *Bachelor of International Business* *2015 – 2019*

Applied AI/Robotics Work: Python, C++, NumPy, TensorFlow, OpenCV, YOLO, ROS, Gazebo, AirSim

- Developed Vietnamese ID card detection and segmentation models using **YOLO** and OpenCV; researched autonomous drone navigation integrating model-predicted heuristic maps with **A* path planning** in AirSim and ROS/Gazebo simulations.
- Completed Udacity Nanodegrees in Robotics and Flying Car Engineering, mastering motion planning, quadcopter PID control, sensor fusion, and robotic arm inverse kinematics.

Applied Mathematics: Linear Algebra, Convex Optimization, Probability & Statistics
Languages: English (Professional Passive Proficiency, Active Skills in Progress)